



UGANDA CHRISTIAN
UNIVERSITY

A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER TEST 2

PROGRAM: *BACHELOR OF SCIENCE IN DATA SCIENCE*

YEAR: 2

SEMESTER: 2

COURSE CODE: *DSC2206*

COURSE NAME: *TIME SERIES ANALYSIS AND FORECASTING*

EXAMINATION TYPE: *PROJECT-BASED TEST*

PROJECT DURATION: MARCH 2026

TIME ALLOWED: *[14 DAYS]*

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This test consists of a project to be executed in [14] days.
 - i. There are two parts to this project-based exam; Part A and Part B.
 - ii. **Part A has Three (3) thematic goals each carrying 25 Marks.** They are all compulsory
 - iii. **Part B has a compulsory task carrying 25 Marks.**
 - iv. **Data Sources:** You are to generate primary datasets or simulated data that suits either of the following seven thematic areas: Health & Biology, Agriculture, Finance, Climate, Education, Transport & Logistics, Internet of Things (IoT) & Manufacturing. The dataset should have no less than 500 observations.
3. **Submission Format:** Submit your project results as a PDF with link to the code solutions in either your **Github** or **Kaggle** account for **Part A** and as a **powerpoint presentation** for **Part B**. **All submissions are to be made on Moodle.**
4. Every student group assigned has a responsibility to prove their contribution towards every milestone, and marks will be awarded to each group.

Project-based assessment guidelines for 7 student groups

S/N	Milestone Description	Maximum Marks
1	MILESTONE ONE Exploratory analysis and data preprocessing	25 %
2	MILESTONE TWO Statistical Diagnostics	25%
3	MILESTONE THREE Model development and Smoothing	25 %
4	MILESTONE FOUR Evaluation and Reporting	25%

TOTAL MARKS 100 %

PART A: [TOTAL 75 MARKS]

Data Sources: You are to generate primary datasets or simulated data that suits either of the following seven thematic areas: Health & Biology, Agriculture, Finance, Climate, Education, Transport & Logistics, Internet of Things (IoT) & Manufacturing. The dataset should have no less than 500 observations.

Goal 1: Exploring and Preprocessing the Dataset [25 Marks]

1. Describe the type of data you're using and discuss the frequency of observation.
2. **Decomposition:** Decompose the series to identify the Trend, Seasonal, and Irregular components. Explain the logic behind choosing an Additive vs. Multiplicative model.
3. **Pre-processing:** If the series shows non-constant variance (heteroscedasticity) or a trend, apply transformations. Document the effect of these transformations on the series' visual properties.

Goal 2: Statistical Diagnostics [25 Marks]

1. **Stationarity:** Perform a formal test to determine if the series is stationary. Explain the null and alternative hypotheses in this context.
2. **Correlation Analysis:** Present and interpret the ACF and PACF plots.
3. **Differencing:** If the series was non-stationary, demonstrate how first or second-order differencing stabilized the mean. Compare the ACF plots of the raw data versus the differenced data to illustrate the removal of trend.

Goal 3: Smooth Data and Develop a Suitable Model. [25 Marks]

1. **Smoothing:** Apply a Simple Exponential Smoothing (SES) or Holt-Winters method. Discuss how the smoothing parameters (alpha, beta, gamma) impact the model's sensitivity to recent versus historical observations.
2. **Model Selection:** Develop at least two competitive models (e.g., ARIMA vs. SARIMA, or Exponential Smoothing vs. ARIMA). Explain the criteria used for selection (e.g., AIC, BIC, or adjusted R^2):
3. **Parsimony:** Justify your final model selection, explaining why it balances predictive power and model complexity

PART B: [TOTAL 25 MARKS]

Goal 4: Model Evaluation and Reporting [25 Marks]

1. **Forecasting:** Generate out-of-sample forecasts for at least 10 steps ahead. Provide a confidence interval for your forecast.
2. **Evaluation:** Evaluate the model performance using appropriate metrics: **MAE, RMSE, and MAPE**. Discuss what these metrics reveal about the model's accuracy.
3. **Reflection:** Discuss the limitations of your model.